

Python Loops - Beginner Guide (Hinglish)

Introduction

Loop ka matlab hai kisi kaam ko baar-baar repeat karna bina same code ko multiple times likhe. Python me mainly 2 loops hote hain: for loop while loop Loops programming ko simple aur efficient banate hain.

1. for Loop

for loop tab use hota hai jab hume pata ho ki loop kitni baar chalana hai.

Syntax:

```
for variable in sequence:  
    statement
```

Example:

```
for i in range(5):  
    print(i)
```

Output: 0 1 2 3 4

range() Function

range() number sequence generate karta hai. range(5) → 0 se 4 range(1, 6) → 1 se 5 range(1, 10, 2) → 1, 3, 5, 7, 9

2. while Loop

while loop tab use hota hai jab tak condition True rahe.

Syntax:

```
while condition:  
    statement
```

Example:

```
i = 1  
while i <= 5:  
    print(i)  
    i += 1
```

3. Nested Loop

Jab ek loop ke andar doosra loop hota hai use nested loop kehte hain.

Syntax:

```
for i in range(3):  
    for j in range(2):  
        print(i, j)
```

Loop Control Statements **break** - Loop ko turant stop karta hai. **continue** - Current iteration skip karke next iteration par jata hai. **pass** - Placeholder statement, kuch nahi karta.

break Example

```
for i in range(10):  
    if i == 5:  
        break
```

```
print(i)
```

continue Example

```
for i in range(5):  
    if i == 2:  
        continue  
    print(i)
```

Beginner Tips Jab repeat count pata ho to for loop use karo. Jab condition ke basis par loop chalana ho to while loop use karo. Infinite loop se bachne ke liye while loop me condition update karo. Nested loops pattern printing aur tables me useful hote hain.